

Kenny Nguyen

US Citizen - Gilbert, Arizona

(480)-939-0964 | kennytrung0603@icloud.com | [linkedin.com/in/kenny-nguyenn](https://www.linkedin.com/in/kenny-nguyenn) | github.com/Intro0 | kennynguyen.me

EDUCATION

Arizona State University

Tempe, AZ

Master of Science in Computer Science, Accelerated 4+1 Program

Aug 2026 – May 2027

Bachelor of Science in Computer Science, GPA: 3.88/4.0

Aug 2022 – May 2026

Relevant Coursework: Foundations of Algorithms, Operating Systems, Computer Networks, Data Structures and Algorithms, Software QA and Testing, Principles of Programming Languages

EXPERIENCE

Opportunity Hack

Remote

Software Engineer Intern

Jun 2025 – Aug 2025

- Led a 3-person team to design and ship a full-stack alumni platform serving **1,000+ users** across a global community, making key architectural decisions including technology selection (Next.js, Supabase, Clerk) and system design.
- Designed the platform's data model and access control architecture, choosing **Row Level Security (RLS)** to enforce admin and user role separation and **Clerk** for authentication with webhook-based account linking.
- Delivered an interactive alumni world map displaying **400+ members** with real-time filtering, a CSV-based bulk import tool that migrated **1,000+ legacy records**, and an admin dashboard for managing the alumni database.
- Coordinated integration of third-party services including **Mapbox** for geocoding, **Supabase** for the PostgreSQL backend, and **Mailgun** for targeted bulk-email campaigns to the user base.

PROJECTS

Redis Clone | *Go, TCP Sockets, Concurrency, RESP Protocol*

- Implemented a Redis server from scratch in **Go** supporting **9 commands** including key-value storage with TTL expiration and **Redis Streams** with blocking reads, range queries, and auto-generated entry IDs by parsing the **RESP protocol** over raw TCP.
- Handled **concurrent client connections** using a goroutine-per-client model with **mutex-protected shared storage**, serving multiple simultaneous clients on port 6379 with no external dependencies.

Shell Clone | *Go, Systems Programming, Process Management*

- Developed a **POSIX-like shell** in Go with a REPL loop, **3 builtin commands**, and **PATH resolution** that searches system directories and validates executable permissions before spawning child processes with full I/O passthrough.

Distributed Storage System | *Python, UDP Sockets, Multi-threading, XOR Parity*

- Designed a **3-process distributed storage system (2,198 lines of Python)** with a manager coordinator, user client, and multi-threaded disk nodes communicating over **UDP sockets** via **23 command handlers**.
- Implemented **XOR-based parity** with rotating parity blocks across N disks, enabling automatic single-disk failure recovery by reconstructing lost data from N-1 surviving blocks using **parallel threaded I/O**.

Compiler | *C++, Recursive Descent Parsing, Intermediate Code Generation*

- Built a compiler in **C++** with a **recursive descent parser** that directly generates a linked-list intermediate representation supporting **6 instruction types** (ASSIGN, CJMP, JMP, IN, OUT, NOOP) for assignments, conditionals, loops, and switch statements.
- Implemented a PC-based execution engine that interprets the generated IR, using conditional and unconditional jumps for control flow and a global memory array with **symbol table management** for variable and constant storage.

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, JavaScript, SQL

Frameworks & Tools: React, Git, Linux/Unix

CERTIFICATIONS

CodePath Intermediate Technical Interview Prep (Advanced), Summer 2025